phonghoang at StanceEval-2026: A Two-Stage Framework for Arabic Stance Detection via Multi-Task Regularization and LLM-Guided Data Augmentation

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Abstract

This paper describes the system submitted by the *phonghoang* team for Track 1 (Seen Targets) of the StanceEval-2026 shared task on Arabic stance detection, where models are trained on provided targets and evaluated on a related target within the same thematic domain. We propose a two-stage framework to address this within-domain target shift. In the first stage, we fine-tune a Multi-Task MARBERT model that jointly learns stance, sentiment, and sarcasm, further regularized with R-Drop and Stochastic Weight Averaging (SWA), achieving F_{avg2} of 0.8561 on the development set. In the second stage, we adapt to the test target ("Women Driving") by combining contrastive few-shot pseudo-labeling via a Large Language Model, fusion with an external Arabic stance dataset, and high-confidence samples filtered from a thematically related training target. The resulting augmented training set is used to adapt our optimal Stage 1 architecture, achieving an official F_{avg2} of 0.8431 on the test set.

1 Introduction

Stance detection in Arabic social media is a challenging natural language processing task that requires determining whether a given text expresses *Favor*, *Against*, or *None* toward a specific target. Arabic poses unique difficulties due to its rich morphology, widespread use of dialectal varieties such as Khaleeji, and the prevalence of implicit sarcasm in online discourse. Track 1 of the StanceEval-2026 shared task (Albalawi et al., 2026a) evaluates how well systems trained on provided targets generalize to a related target within the same thematic domain.

To address these challenges, we propose a two-stage pipeline. In the first stage, targeting the provided training targets, we build a multi-task learning framework (Caruana, 1997) on top of MARBERTv2 (Abdul-Mageed et al., 2021), jointly training stance detection alongside auxiliary sentiment and sarcasm classification. We further apply R-Drop regularization (Liang et al., 2021) and Stochastic Weight Averaging (SWA) (Izmailov et al., 2019) to improve model stability and generalization.

In the second stage, we address the within-domain target shift introduced by the test set ("Women Driving"). Rather than modifying the model architecture, we focus on data adaptation: we filter high-confidence samples from a thematically related training target, incorporate external data from ArabicStanceX (Alkhathlan et al., 2025), and generate pseudo-labels using contrastive few-shot prompting with Gemini 3.6 Flash (Team et al., 2025) to construct an augmented training set (Lee, 2013).

Our results show that architectural regularization (R-Drop, SWA) effectively reduces overfitting on limited training data, while target-wise threshold tuning and data fusion are the main contributing factors for cross-target generalization on the test set. To facilitate reproducibility, our code and experimental pipelines are publicly available. [1]

2 Background

2.1 Task Setup and Dataset

StanceEval-2026 Track 1 (Albalawi et al., 2026a) requires classifying the stance of an Arabic tweet toward a given target into one of three categories: *Favor*, *Against*, or *None*. Training and development data are drawn from the Mawqif-XT dataset (Albalawi et al., 2026b), containing tweets about three targets: COVID-19 Vaccine, Digital Transformation, and Women Empowerment. A summary of the targets across the dataset splits is provided in Table 1. Beyond stance labels, the dataset also provides per-tweet annotations for sentiment (Positive,

[1] https://github.com/phonghoangKHDL/phonghoang-StanceEval-2026

Negative, Neutral) and sarcasm (Yes, No).

The track evaluates generalization to a related target within the same thematic domain. Specifically, the test set introduces the target "Women Driving", which shares a semantic relationship with the training target "Women Empowerment".

Table 1: Summary of targets and sample sizes across Mawqif-XT splits.

Split	Target	Samples
Train	COVID-19 Vaccine	1,167
	Digital Transformation	1,145
	Women Empowerment	1,190
Dev	COVID-19 Vaccine	206
	Digital Transformation	203
	Women Empowerment	210
Test (Track 1)	Women Driving	352

2.2 Related Work

Arabic stance detection relies heavily on transformer-based models pretrained on large Arabic corpora. MARBERT (Abdul-Mageed et al., 2021) and AraBERT (Antoun et al.) have become standard backbones for Arabic NLP tasks due to their coverage of both Modern Standard Arabic and dialectal varieties.

Cross-target stance detection — where the test target differs from those seen during training — remains a known challenge in the field. Prior work has addressed this through data augmentation and pseudo-labeling (Lee, 2013), as well as multi-task learning (Caruana, 1997) to leverage auxiliary signals. More recently, large language models have been applied for few-shot stance annotation, reducing the need for costly manual labeling in low-resource target settings. Our work builds on these directions by combining multi-task regularization with LLM-guided data augmentation and external dataset fusion via ArabicStanceX (Alkhathlan et al., 2025).

3 System Overview

Figure 1 illustrates our overall two-stage methodology for in-domain optimization and cross-target adaptation.

3.1 Data-Centric Adaptation Pipeline

To adapt the model to the test target ("Women Driving"), we constructed an augmented training set of 2,220 samples by combining three data sources:

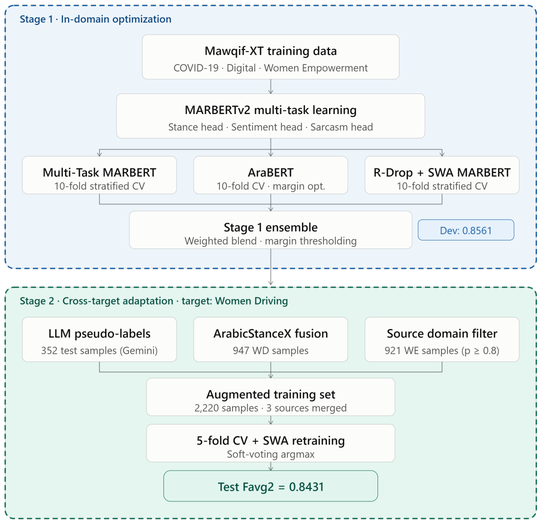

Figure 1: Overview of our two-stage framework.

- **LLM Pseudo-Labeling:** We used Gemini Flash (Team et al., 2025) to generate pseudo-labels for the 352 test samples (Lee, 2013). To handle implicit Khaleeji sarcasm, we designed a contrastive few-shot prompt that explicitly contrasts genuine support with sarcastic mockery.

- **External Data Fusion:** We extracted 947 tweets about "Women Driving" from the ArabicStanceX dataset[2] (Alkhathlan et al., 2025).

- **Source Domain Filtering:** We selected 921 high-confidence samples (confidence ≥ 0.8) from the "Women Empowerment" target in the original Mawqif-XT training and development sets, leveraging their thematic proximity to the test target.

3.2 Multi-Task MARBERT Architecture

We base our system on MARBERTv2 (Abdul-Mageed et al., 2021), appending three parallel classification heads for stance, sentiment, and sarcasm prediction in a multi-task learning framework (Caruana, 1997). All three heads share the same `[CLS]` token representation as input.

For the primary stance task, we apply weighted cross-entropy loss with label smoothing of 0.1. Class weights are computed automatically from the training label distribution using balanced weighting to address class imbalance. The auxiliary sentiment and sarcasm heads use standard cross-entropy

[2] https://huggingface.co/datasets/Faris-ML/ArabicStanceX

loss, with missing labels from external and pseudo-labeled data masked via `ignore_index=-100`.

3.3 Regularization and Ensembling

To reduce overfitting, we apply R-Drop regularization (Liang et al., 2021), which encourages consistency between the output distributions of two stochastic forward passes of the same input. With a dropout rate of $p = 0.2$, the symmetric KL divergence between the two predictions is added to the training objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + 0.5 \times \mathcal{D}_{\text{KL}}^{\text{sym}}(P_1 \parallel P_2)$$

Our cross-validation strategy differs between the two stages. In the first stage, with 3,502 training samples, we use 10-Fold Stratified Cross-Validation. In the second stage, given the smaller augmented dataset (2,220 samples) and the presence of noisy pseudo-labels, we reduce this to 5-Fold to avoid overfitting to narrow validation splits.

During the development phase, an AraBERT-based ensemble (0.6 MARBERT, 0.4 AraBERT) was evaluated to encourage architectural diversity. However, due to its inferior performance (table 2), our final Stage 1 ensemble relies exclusively on MARBERT variants.

For the final development ensemble, rather than a standard soft-voting approach, we constructed a weighted blend of our strongest independent pipelines. Specifically, the final predictions were generated by weighting the 10-Fold Multi-Task MARBERT probabilities by a factor of 2.0 and the 10-Fold R-Drop+SWA MARBERT probabilities by 1.0. This 2:1 ratio was empirically chosen to favor the slightly higher individual validation performance of the Multi-Task configuration while still benefiting from the regularization and stability provided by the SWA models.

3.4 Inference and Decision Making

Rather than applying standard `argmax` to the model probabilities, we use different decision strategies for each stage:

- **Target-wise Margin Thresholding (Stage 1):** Since label distributions differ across targets, we tune per-target decision thresholds $(\theta_{\text{Against}}, \theta_{\text{Favor}})$ using Out-Of-Fold (OOF) probabilities. A sample is assigned *None* only if neither *Against* nor *Favor* probabilities exceed their respective thresholds.

Table 2: Ablation results on the development set.

Model Configuration	Dev F_{avg2}
MARBERT (Focal Loss + Threshold Tuning)	0.8302
MARBERT (Multi-Task Learning)	0.8250
AraBERT (10-Fold CV + Margin)	0.7923
MARBERT+AraBERT Ensemble	0.8366
MARBERT (10-Fold + Multi-Task)	0.8527
MARBERT (10-Fold + R-Drop + SWA)	0.8501
Final Dev Ensemble (2:1 Weighted Blend)	**0.8561**

- **Soft-Voting Argmax (Stage 2):** For the test set submission, we average the softmax outputs across all 5 SWA fold models and apply `argmax` on the result, reducing sensitivity to individual model predictions.

4 Experimental Setup

4.1 Preprocessing and Hyperparameters

Our preprocessing normalizes Arabic-specific characters by mapping Alef variants to , Yeh to , and Teh Marbuta to . We also remove URLs, user mentions, Tatweel characters, and consecutive repeated letters. Text is tokenized using the MARBERTv2 tokenizer with a maximum sequence length of 128.

All models were trained for 6 epochs using AdamW with a learning rate of 2×10^{-5}, a batch size of 16, a cosine learning rate scheduler, a warmup ratio of 0.1, and a weight decay of 0.01.

4.2 Evaluation Metric

Following the official evaluation protocol (Albalawi et al., 2026a), the primary metric is F_{avg2}, defined as the macro-averaged F1-score over the *Favor* and *Against* classes only:

$$F_{\text{avg2}} = \frac{F1_{\text{Favor}} + F1_{\text{Against}}}{2}$$

The *None* class is excluded from this computation. All threshold tuning in our experiments targets this metric directly.

5 Results and Analysis

5.1 Quantitative Results and Ablation

Stage 1: In-Domain Optimization. For the single-model Multi-Task baseline (Table 2), we report the reproducible saved checkpoint score (0.8250), though peak intermediate validation reached 0.8282. Combining MARBERT with AraBERT in a weighted ensemble yields only a modest gain (0.8366), even underperforming the

Table 3: Stage 2 ablation on the official test set (target: *Women Driving*).

Stage 2 Configuration	Test F_{avg2}
Stage 1 ensemble (no adaptation)	0.6701
+ LLM few-shot annotation	0.8310
+ LLM pseudo-labeling	0.8382
+ External data fusion (ArabicStanceX)	**0.8431**

MARBERT-only 10-Fold configuration (0.8527). This gap can be explained by the performance disparity between the two backbones: AraBERT reaches only 0.7923 on the development set, and its lower-quality predictions partially offset the stronger MARBERT signal. This motivated our decision to build the Stage 1 ensemble around MARBERT variants only. Combining 10-Fold Cross-Validation, multi-task learning, R-Drop, and SWA, we reach a peak development score of 0.8561.

Stage 2: Cross-Target Adaptation. Without any adaptation, the Stage 1 ensemble drops sharply to 0.6701 on the official test set (Table 3), confirming the difficulty of the target shift from "Women Empowerment" to "Women Driving". Adding LLM few-shot annotation alone recovers most of this gap (+16.1 pp), suggesting that the contrastive few-shot prompt helps the model handle target-specific sarcasm patterns. Retraining with these pseudo-labels adds a further +0.7 pp, and fusing external data from ArabicStanceX brings the final score to 0.8431 (+0.5 pp).

Cross-Target Gap Analysis. The 1.3 pp gap between the development score (0.8561) and the official test score (0.8431) is expected. The development set covers the same three training targets, while the test set introduces a related but lexically and stylistically distinct target. As shown by the baseline drop to 0.6701 in Table 3, this gap reflects the inherent difficulty of cross-target generalization rather than model instability.

5.2 Error Analysis

A manual inspection of misclassified samples shows that implicit sarcasm in Khaleeji Arabic is one of the main challenges for baseline models. In the context of the test target ("Women Driving"), tweets using humor or positive expressions are often misclassified as *Favor*. For example, tweets that mimic genuine support while actually mocking women's driving ability were frequently misunderstood. Our Stage 2 contrastive few-shot prompt was designed to make this distinction explicit, helping generate more accurate pseudo-labels for such cases.

6 Limitations

While our system demonstrates strong cross-target generalization, several limitations remain. First, our data augmentation pipeline relies exclusively on a single proprietary Large Language Model (Gemini 3.6 Flash) for pseudo-labeling. Second, our explicit use of pseudo-labeled test instances and the integration of external target-specific data (ArabicStanceX for "Women Driving") fundamentally alters the training distribution. We acknowledge that this data-centric augmentation may impact the direct comparability of our results with teams strictly adhering to the initially provided dataset constraints. Finally, our adaptation methodology was explicitly optimized for the thematic domain shift from "Women Empowerment" to "Women Driving"; its efficacy on entirely disparate, strict zero-shot topics requires further empirical validation.

7 Conclusion

We proposed a two-stage framework for Arabic stance detection, achieving robust in-domain performance via multi-task regularization and effective cross-target adaptation through LLM pseudo-labeling and external data fusion. Future work will investigate more reliable pseudo-labeling strategies for low-resource target transfer.

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A LLM Prompt for Pseudo-Labeling

To generate pseudo-labels for the zero-shot target (“Women Driving”), we utilized Gemini 3.6 Flash with a carefully crafted contrastive few-shot prompt. The prompt includes specific rules to handle Khaleeji sarcasm and religious/social objections. The exact prompt template is provided below:

System Prompt:
You are an elite expert in Saudi sociolinguistics, Khaleeji dialects, and Twitter sentiment analysis.

Classify the EXACT stance of the following tweet regarding the topic: ’Women Driving’.

RULES:

1. Output ONLY ONE WORD: ’Favor’, ’Against’, or ’None’.
2. SARCASM ALERT: Saudi Twitter frequently uses sarcasm. If a tweet seemingly praises women driving but actually mocks their driving skills, predicts accidents, or uses laughing emojis ironically, it is strictly ’Against’.
3. RELIGIOUS/SOCIAL OBJECTION: Claims that women driving causes corruption or mixes genders is strictly ’Against’.
4. GENUINE SUPPORT: Joy, relief, personal milestones, or defending the royal decree is ’Favor’.

CONTRASTIVE EXAMPLES:

[Example Pair 1: Sarcasm vs. Genuine Joy]
Tweet A: وأخيراً بنسوق سياراتنا ونفتك من ذل السواقين شكرا لملكنا
Stance A: Favor (Genuine relief and gratitude)
Tweet B: إي خلهم يسوقون عشان نشوف حوادث تضحك في الشوارع ونوسع صدورنا [laughing emoji]
Stance B: Against (Sarcastic, mocking their skills, anticipating accidents)

[Example Pair 2: Religious/Social Fear vs. Neutral News]
Tweet A: المرأة مكانها بيتها القيادة بتفتح أبواب الفساد والاختلاط
Stance A: Against (Ideological opposition)
Tweet B: أعلن المرور السعودي عن بدء استبدال الرخص الدولية برخص سعودية للنساء
Stance B: None (Objective news reporting)

[Example Pair 3: Implicit Mocking]
Tweet A: جهزوا التأمين الشامل ياشباب، الشوارع بتصير ملاهي سيارات تصادم
Stance A: Against (Implicitly mocking women as bad drivers using humor)

[Example Pair 4: Personal Story — Genuine Favor]
Tweet A: اليوم أول مرة أوصل أولادي المدرسة بسيارتي، شعور ما يوصف، أخيراً حرة
Stance A: Favor (Personal milestone, genuine emotion of freedom and pride)

Input:
Now, deeply analyze and classify this new tweet:
Tweet: {tweet_text}
Stance: